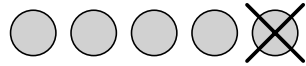
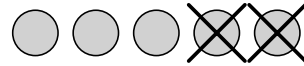


Two players share one pile of counters. They take turns. On your turn, take away 1 counter or 2 counters. Whoever takes the last counter wins.



take 1



take 2

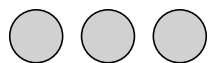
Problem 1: It is your turn. Can you take the last counter on this turn? Circle yes or no under each pile.


 yes no


 yes no


 yes no

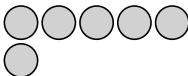
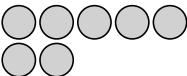
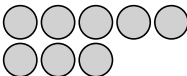
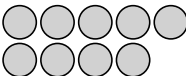
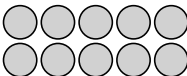
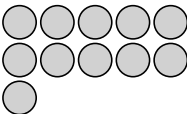
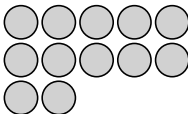
Problem 2: There are 3 counters, and it is your turn. Your partner always plays carefully. Can you be the player who takes the last counter? Circle yes or no.



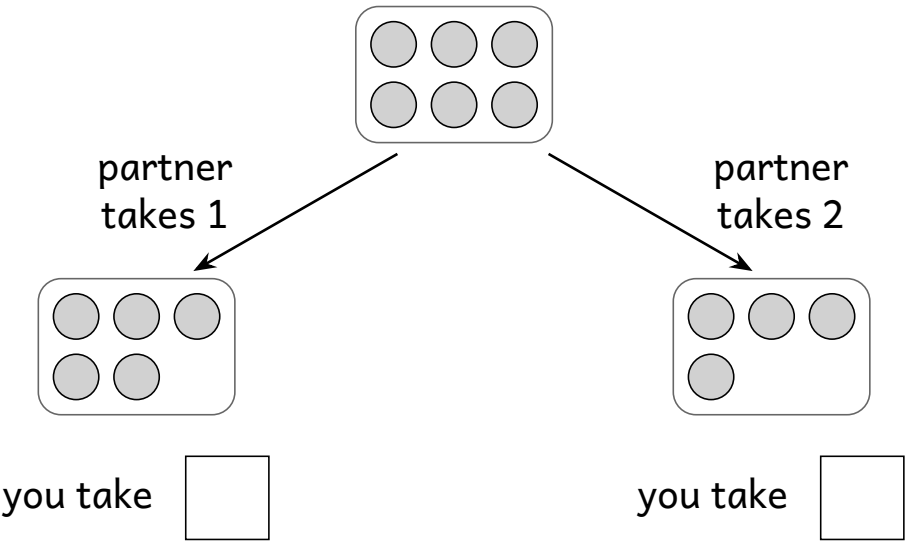
yes
no

A pile is a **trap** when the player whose turn it is will lose if the other player plays carefully.

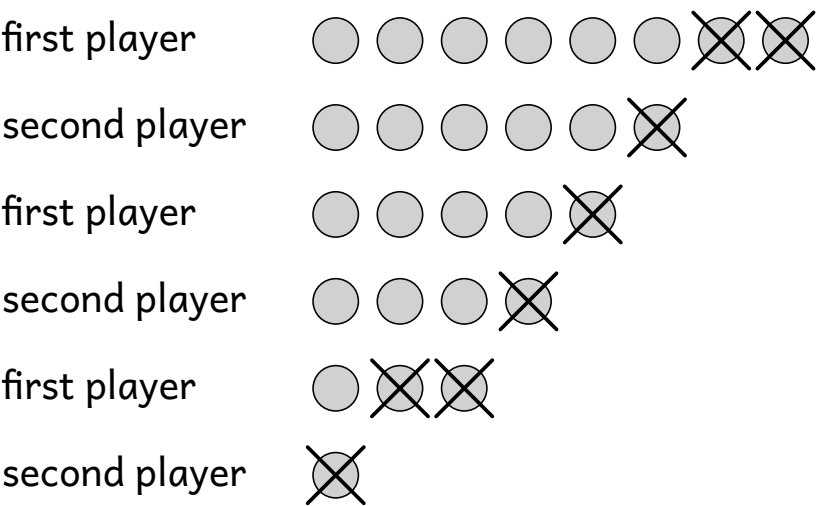
Problem 3: Which of these piles are traps? Draw an X in the box under each trap.

1  ☐	2  ☐	3  ☐	4  ☐	5  ☐	6  ☐
7  ☐	8  ☐	9  ☐	10  ☐	11  ☐	12  ☐

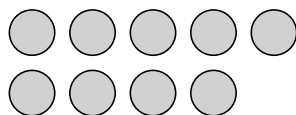
Problem 4: There are 6 counters, and it is your partner’s turn. Your partner might take 1 or might take 2. For each choice, write how many counters you will take next so that you are sure to win.



Problem 5: This game started with 8 counters. Each row shows one turn, and the crossed-out counters are the ones taken on that turn. The first player lost. Circle the row where the first player could have taken a different number of counters and then been sure to win.



Problem 6: You will play against the adult at your table with a pile of 9 counters. Before you start, you choose who goes first. Circle the choice that lets you win, however the adult plays.

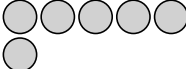
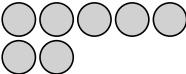
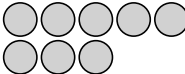
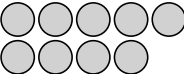
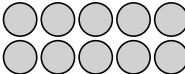
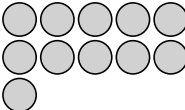
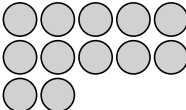
☐ I go first☐ The adult goes first

Problem 7: Find a trap with more than 12 counters and a trap with more than 20 counters. Draw each one or write its number in a box.

more than 12

more than 20

Problem 8: Play with a different rule: on your turn, take away 1, 2, or 3 counters. Whoever takes the last counter wins. Which piles are traps now? Draw an X in the box under each trap.

1 	2 	3 	4 	5 	6 
7 	8 	9 	10 	11 	12 

Problem 9: Now there are two piles. On your turn, choose one pile and take 1 or 2 counters from that pile only. Whoever takes the last counter wins. For each game, circle whether you would rather go first or go second.

		go first	go second
		go first	go second
		go first	go second
		go first	go second
		go first	go second